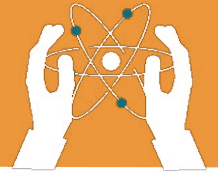


UNIT 3

TRANSPORT IN LIVING ORGANISMS

Learning objectives:

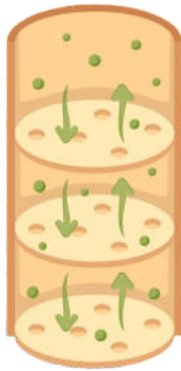
- Understand the role of transport in maintaining homeostasis.
- Learn how water and nutrients are transported in plants.
- Identify the components of blood and the function of the heart and blood vessels.



3.1 Introduction to Transport

Importance of Transport in Living Organisms

Transport is the movement of essential substances like water, nutrients, and waste within an organism. It is necessary for cell function and maintaining homeostasis. Both plants and animals rely on transport systems for survival and proper function.

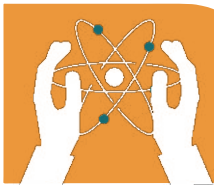


3.2 Transport in Plants

Water and Ion Uptake (Structure and Function of Root Hairs)

Root hairs are tiny structures on plant roots that absorb water and essential nutrients from the soil. They increase the surface area for





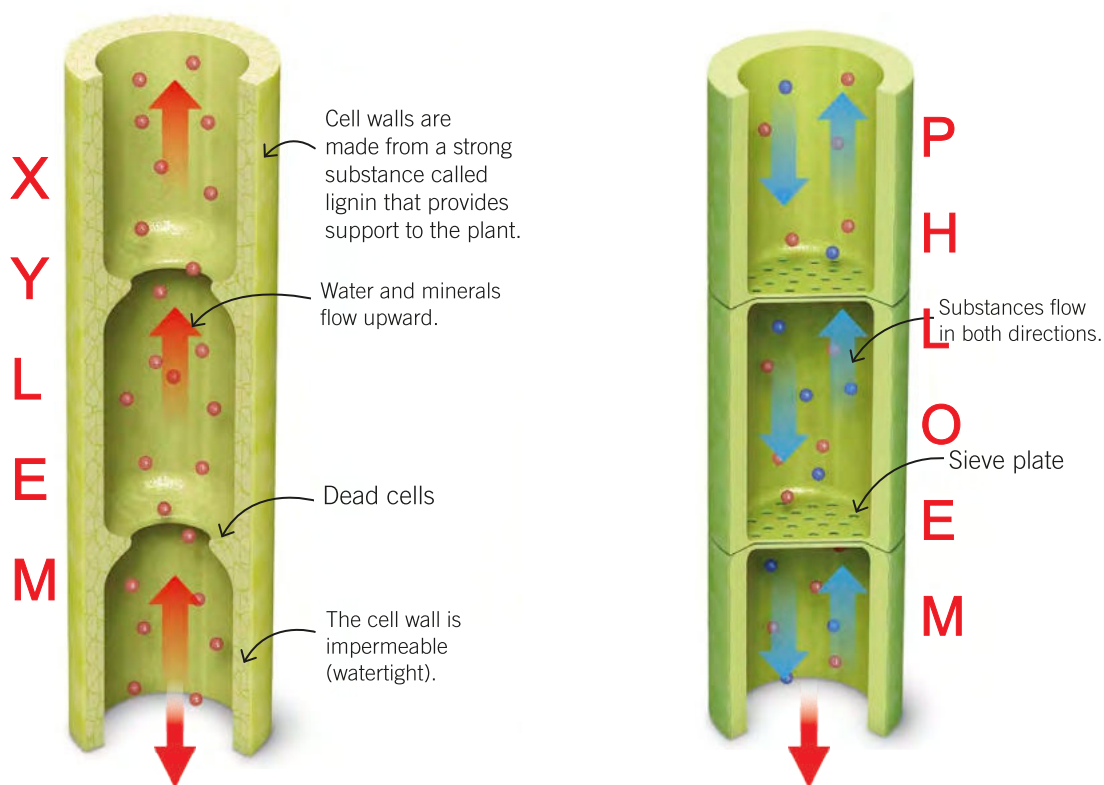
TRANSPORT IN LIVING ORGANISMS

Transpiration: Introduction and Significance

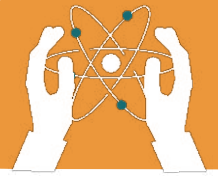
Transpiration is the process where water evaporates from the leaves, pulling more water and nutrients from the roots. This helps maintain plant health and regulates temperature.

Pathway of Water and Food in the Stem

The stem and roots of the plant have specialized parts called xylem and phloem to transport material. Water and nutrients travel through the xylem from roots to leaves, while food (sugars) produced in the leaves during photosynthesis is transported through the phloem to other parts of the plant.



Xylem carries water and minerals from roots to leaves.	Phloem carries food materials from leaves to the other parts.
It is composed of thick dead cells.	It is composed of living cells.
It has a thick cell wall.	It has a thin cell wall.
It gives mechanical strength to the plants.	It doesn't gives mechanical strength to the plants.

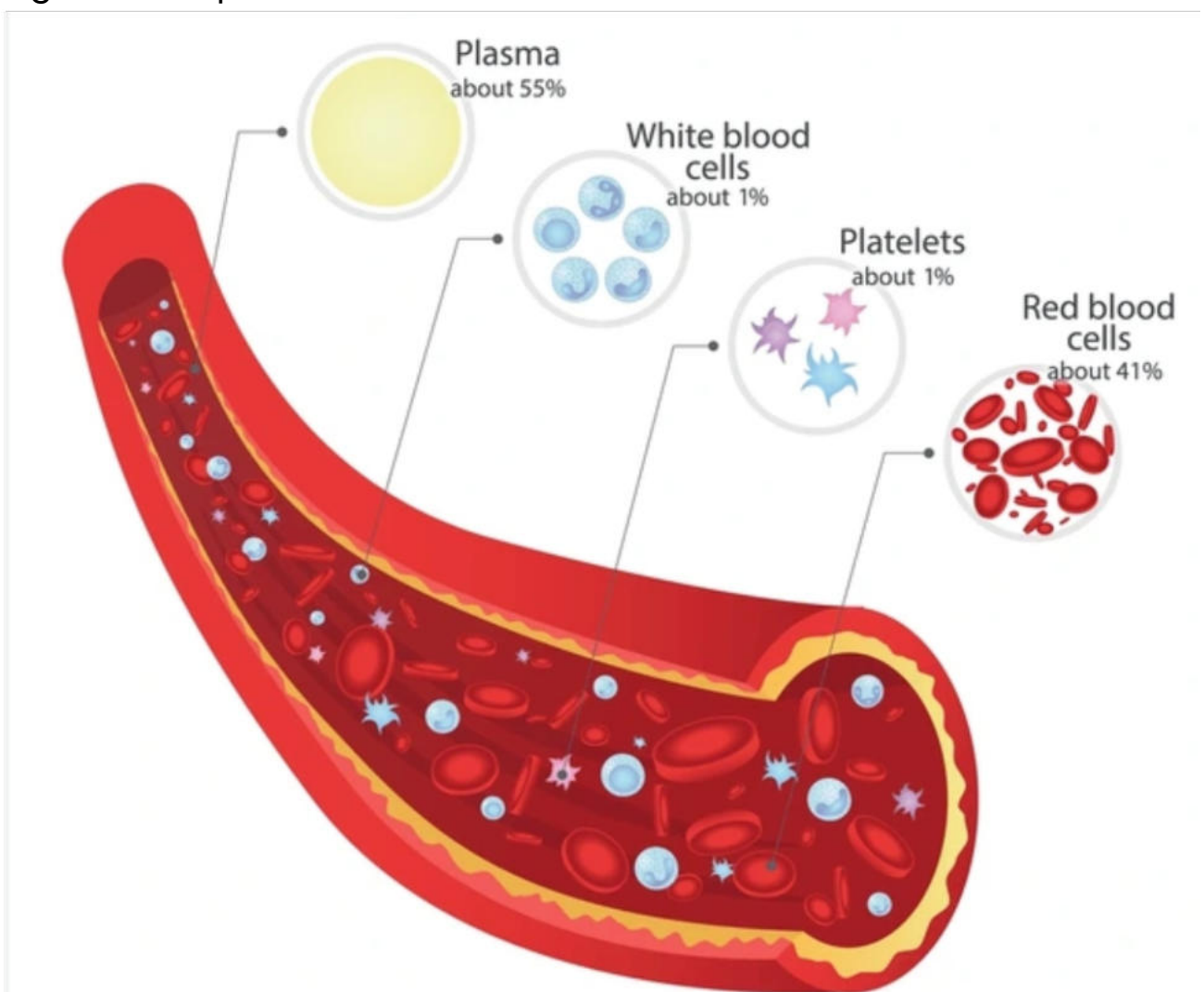


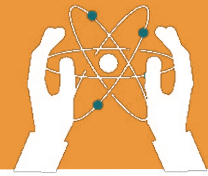
3.3 Transport in Animals

Blood and Its Components

Blood transports oxygen, nutrients, and waste. It has four components:

- **Red Blood Cells (RBCs):** They carry oxygen using haemoglobin.
- **White Blood Cells (WBCs):** They fight against infections.
- **Platelets:** They help in clotting.
- **Plasma:** It transports nutrients, hormones, and waste, and helps regulate temperature.



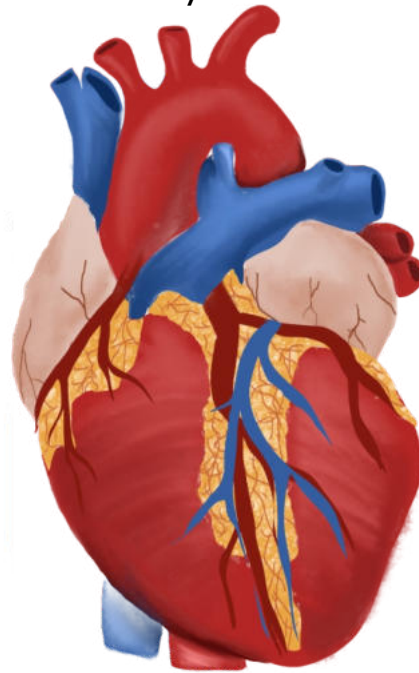
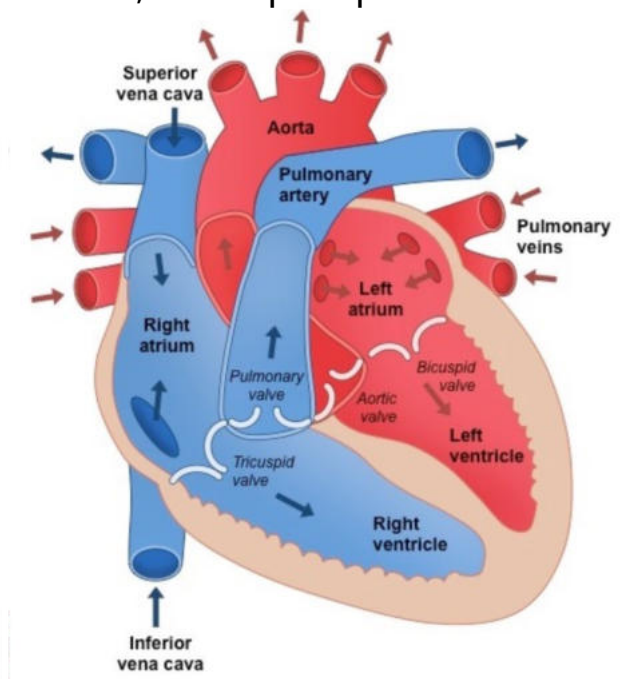


3.4 Human Heart

Structure and Role of the Heart

The human heart is a muscular organ that pumps blood throughout the body. It has four chambers: two atria (upper chambers) and two ventricles (lower chambers). Blood flows through the heart in a specific sequence:

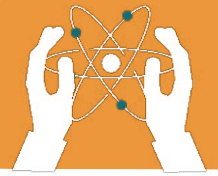
- From the body into the right atrium, then to the right ventricle, which pumps it to the lungs for oxygenation.
- The oxygenated blood returns to the left atrium, flows into the left ventricle, and is pumped to the rest of the body.



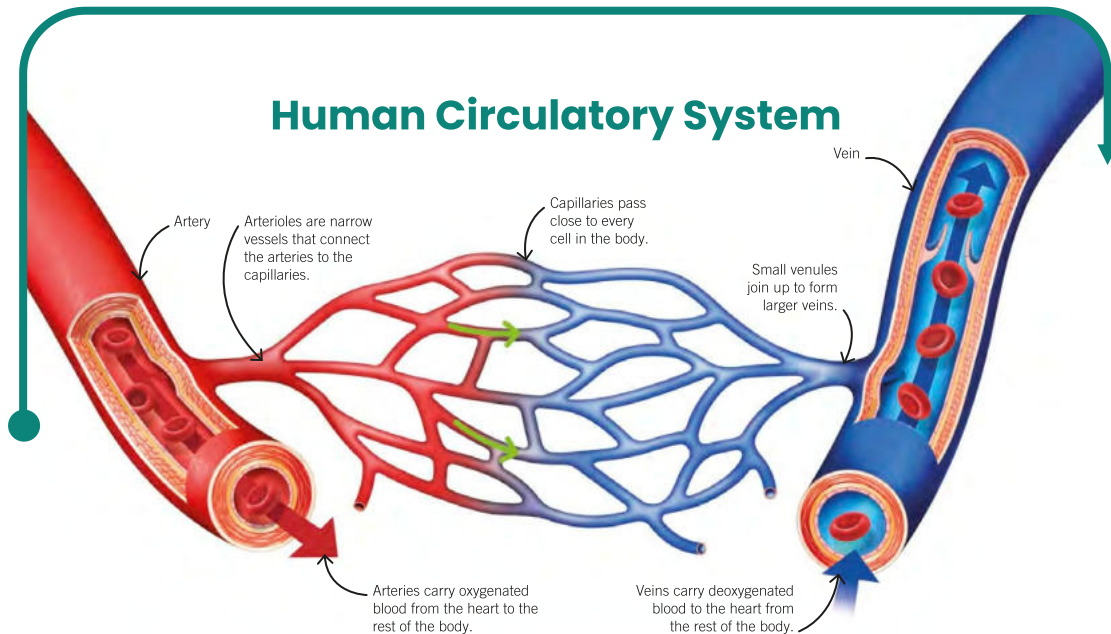
• Blood Vessels

Blood vessels are tubes that carry blood throughout the body. There are three types of blood vessels:

- **Arteries:** They carry oxygenated blood away from the heart to the body.
- **Veins:** They carry deoxygenated blood from the body back to the heart.



- **Capillaries:** These are tiny vessels where the exchange of oxygen, nutrients, and waste products occurs between the blood and tissues.



DISORDERS IN HUMAN CIRCULATORY SYSTEM

✦ **Heart Attack:**

Sometimes, the heart gets sick, and this is called a "heart attack." It happens when the heart doesn't get enough blood. It can be because of unhealthy food or not exercising.

✦ **High Blood Pressure:**

"High blood pressure" means the blood in your body is pushing too hard against the walls of your blood vessels. It's like blowing up a balloon too much. It can happen if you eat too much salty food.

✦ **Diabetes:**

"Diabetes" is when your body can't use sugar properly. It's like a problem with the sweet stuff in your blood. You might need to take medicine or eat healthy foods to manage it.

✦ **Leukaemia:**

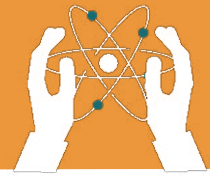
"Leukemia" is a type of blood cancer. It happens when your body makes too many white blood cells.

✦ **Haemophilia:**

"Hemophilia" is when the blood doesn't clot properly. This means when you get a cut, it takes a long time to stop bleeding. It's a rare condition that some people have.

✦ **Anaemia:**

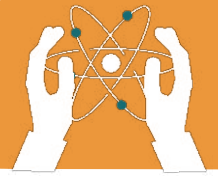
"Anemia" is when you don't have enough red blood cells in your body. Red blood cells help carry oxygen to your body. Eating healthy foods with iron can help prevent it.



EXERCISES:

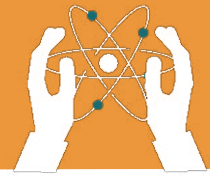
Fill in the blanks.

1. The process of water loss in plants is called _____.
2. The structure that absorbs water and minerals from the soil in plants is called _____.
3. _____ blood cells carry oxygen to different parts of the body.
4. The heart has _____ chambers.
5. Platelets help in _____ when a blood vessel is injured.
6. _____ blood vessels carry blood away from the heart.
7. The main function of red blood cells is to transport _____.
8. The _____ connects the right ventricle to the lungs for oxygenation.



Tick(✓) for True and cross (x) for False statements.

1. Transpiration helps plants absorb water from the soil. ☐
2. White blood cells are responsible for transporting oxygen. ☐
3. Arteries carry oxygenated blood to the body. ☐
4. Blood vessels help in transporting oxygen, nutrients, and waste products. ☐
5. The heart has six chambers. ☐
6. Plasma is responsible for fighting infections. ☐
7. The phloem transports water in plants. ☐
8. Root hairs are responsible for nutrient absorption in plants. ☐



Choose the correct answer.

1. What is the primary function of white blood cells?

- a) To carry oxygen b) To fight infections
- c) To form blood clots d) To transport nutrients

2. What part of the plant absorbs water from the soil?

- a) Leaves b) Roots
- c) Stem d) Flowers

3. Which of the following carries deoxygenated blood back to the heart?

- a) Arteries b) Veins
- c) Capillaries d) Red blood cells

4. Where does the exchange of gases take place in animals?

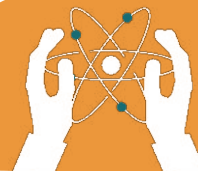
- a) Heart b) Lungs
- c) Blood vessels d) Kidneys

5. What is the function of the stomata in plants?

- a) To absorb water b) To release oxygen
- c) To exchange gases d) To produce food

6. Which part of the human body is responsible for regulating body temperature?

- a) Heart b) Kidneys
- c) Skin d) Lungs



7. What is the main function of platelets in the blood?

- a) To carry oxygen
- b) To fight infections
- c) To help in blood clotting
- d) To transport nutrients

8. Which of the following processes helps regulate water loss in plants?

- a) Photosynthesis
- b) Germination
- c) Transpiration
- d) Respiration

Answer the Following Questions:

Short-Answer Questions.

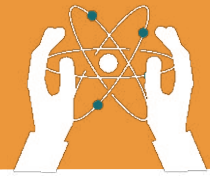
- 1. What is the role of root hairs in plants?**
- 2. How does transpiration benefit plants?**
- 3. Write the difference between arteries and veins.**
- 4. Write the difference between xylem and phloem.**
- 5. Name the three types of blood vessels and describe their functions.**
- 6. How many chambers are there in the human heart?**

Long-Answer Questions.

- 1. Describe the structure of the human heart and explain its role in circulation.**
- 2. Explain how blood components work together to maintain body function.**

Short Note

- 1. Write a short note on “Disorders of the human heart”.**



TRANSPORT IN LIVING ORGANISMS



Creating a Model of Circulatory System



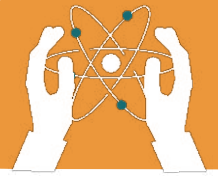
ACTIVITY

Create a model of the human circulatory system using cardboard or other materials. Label the heart, blood vessels, and describe how blood circulates through the body.



Critical Thinking

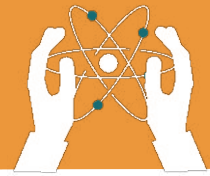
- Why do you think it is important for plants to regulate the loss of water through transpiration? How could this process impact the plant if it did not occur?



1. Name four things plants need for healthy growth.

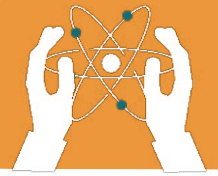
2. Fill in this table to show the properties of xylem and phloem cells.

	Xylem	Phloem
Cell structure		column of living cells connected by tiny holes
Materials transported		
Materials transported from		
Materials transported to	shoots and leaves	
Materials travel in one or both directions?		



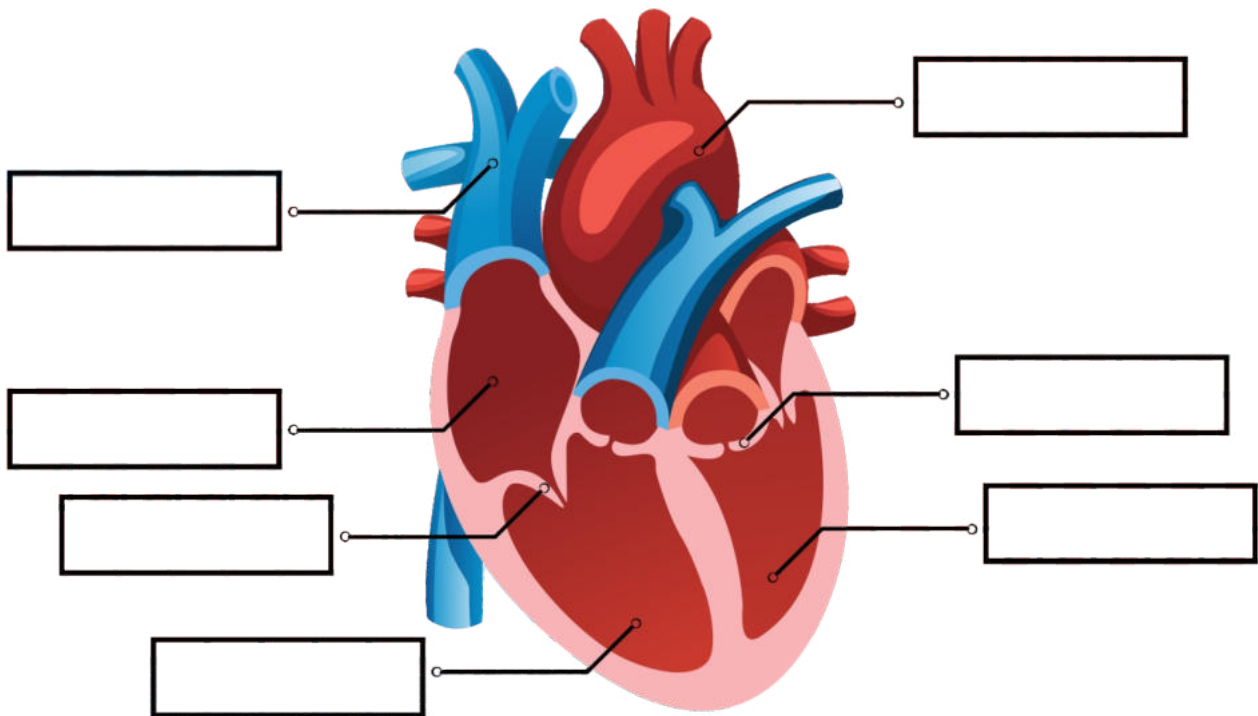
2. Complete the following passage by filling in the blanks.

Transport is the movement of essential substances like _____ (1), _____ (2), and waste within living organisms. It is essential for _____ (3) function and maintaining _____ (4). Both plants and animals rely on transport systems for _____ (5). In plants, root hairs are tiny structures on plant roots that absorb _____ (6) and essential _____ (7) from the soil. These root hairs increase the _____ (8) area for absorption, helping plants stay _____ (9) and receive necessary nutrients. In the process of _____ (10), water evaporates from the leaves, pulling more water and nutrients from the roots. Water and nutrients travel through the _____ (11) in plants, while food produced in the leaves is transported through the _____ (12) to other parts of the plant.



The Human Heart

Drag and Drop The Parts of Human Heart.



Word Bank

Left ventricle

Right atria

Aorta

Mitral valve

Right ventricle

Tricuspid valve

Superior vena
cava